

A15536W1

SECOND PUBLIC EXAMINATION

Honour School of Philosophy, Politics and Economics

Honour School of Economics and Management

Honour School of History and Economics

MICROECONOMIC ANALYSIS

OUTLINE ANSWERS

TRINITY TERM 2018

Thursday 7th June 2018, 09:30 - 12:30

Please start the answer to each question on a separate page.

There are 6 questions in this paper.

*Answer **four** questions.*

For multipart questions, the weight assigned to each part is indicated in square brackets.

Candidates may use their own calculators.

Do not turn over until told that you may do so.

1. (a) (i) We can use the Gaussian technique to obtain

$$\left(\begin{array}{cccc|c} -1 & 3 & 1 & -6 & -5 \\ 2 & 1 & 5 & 5 & 3 \\ 1 & 2 & 4 & 1 & \alpha \end{array} \right) \sim \left(\begin{array}{cccc|c} -1 & 3 & 1 & -6 & -5 \\ 0 & 7 & 7 & -7 & -7 \\ 0 & 5 & 5 & -5 & \alpha - 5 \end{array} \right) \sim$$

$$\left(\begin{array}{cccc|c} -1 & 3 & 1 & -6 & -5 \\ 0 & 7 & 7 & -7 & -7 \\ 0 & 0 & 0 & 0 & \alpha \end{array} \right).$$

It is evident from the Row Echelon form of the matrix formed by vectors u_1, u_2, u_3 and u_4 that the rank of these vectors is equal to 2. Since we need three independent vectors to span $\mathbf{R}^3$, these vectors cannot span the entire space.

- (ii) We know that the vector v_α is spanned by vectors u_1, u_2, u_3 and u_4 if and only if the system of linear equations $Ux = v_\alpha$ has at least one solution. From the extended Row Echelon form, we know that this system has infinite solutions whenever $\alpha = 0$, and zero otherwise. Hence, the answer is positive if and only if $\alpha = 0$.

For the second part, notice that one possible solution is $x_1 = -1, x_2 = x_3 = 0$ and $x_4 = 1$, which means that $u_4 - u_1 = v_0$.

- (iii) We need three independent vectors from this collection. It is straightforward to show that vectors u_1 and u_4 are independent. We also know from (b) that v_1 (but not v_0) is independent of u_1 and u_4 , and hence it can be used to form the basis.

- (b) For point $(0, 0)$, we can apply the definition of partial derivative to get

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^2/h^2 - 1}{h} = 0.$$

$$\text{Similarly, } f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h^2/h^2 - 1}{h} = 0.$$

The function has partial derivatives at the origin.

For any other point, the function is a quotient of polynomials where the denominator is not zero, and hence, partial derivatives always exist.

It is not difficult to see that the function f is not continuous.

Notice, for instance, that the sequence of points $\{(\frac{1}{n}, \frac{1}{n})\}$ converges to the origin.

At each of these points, the function f takes values $f(\frac{1}{n}, \frac{1}{n}) = 0$, but the limit of images should be $f(0, 0) = 1$.

Hence, continuity does not hold.

As a direct implication, the function is neither differentiable nor C^1 .

(Notice that, here, students may also try other approaches using continuity of partial derivatives and the formal definition of differentiability).

Finally, failure of C^1 (also, partial derivatives are zero) does not allow the use of the implicit function theorem at the origin.

2. (a) (i) f is concave: $f(\lambda a + (1 - \lambda)b) \geq \lambda f(a) + (1 - \lambda)f(b)$ for all $\lambda \in (0, 1)$.

When f is (strictly) concave, a local maximum or a stationary point is a (unique) global maximum.

With a concave objective and convex constraints, the Kuhn-Tucker first order conditions are sufficient for a global maximum.

- (ii) • Write $k(x) = f(x) + g(x)$. Then:

$$\begin{aligned} k(\lambda a + (1 - \lambda)b) &= f(\lambda a + (1 - \lambda)b) + g(\lambda a + (1 - \lambda)b) \\ &\geq \lambda f(a) + (1 - \lambda)f(b) + \lambda g(a) + (1 - \lambda)g(b) \\ &= \lambda k(a) + (1 - \lambda)k(b) \end{aligned}$$

• Let $g = 2f$; then $f - g = -f$ which is convex.

• Let $f(x) = g(x) = x^\alpha$ with $\frac{1}{2} < \alpha < 1$; then $fg = x^{2\alpha}$ which is convex.

- (b) (i) $f(x, y) = \alpha \ln x + \beta \ln y$

Since $\alpha > 0$, $\beta > 0$, both $\alpha \ln x$ & $\beta \ln y$ are concave functions of (x, y) (candidates must note this): use the result that the sum of concave functions is concave.

Or examine Hessian:

$$\begin{pmatrix} -\alpha x^{-2} & 0 \\ 0 & -\beta y^{-2} \end{pmatrix}$$

$\alpha > 0$, $\beta > 0$, therefore $\det > 0$, $\text{trace} < 0$, so H is negative definite on $\mathbf{R}_{++}^2$, and f is concave.

- (ii) $g(x, y) = x^\alpha y^\beta$

Hessian:

$$x^{\alpha-2} y^{\beta-2} \begin{pmatrix} \alpha(\alpha-1)y^2 & \alpha\beta xy \\ \alpha\beta xy & \beta(\beta-1)x^2 \end{pmatrix}$$

Want $\det > 0$, $\text{trace} < 0$, so H is negative definite on $\mathbf{R}_{++}^2$, and g is concave.

$$\det =_{\text{sgn}} [\alpha(\alpha-1)\beta(\beta-1)x^2y^2 - \alpha^2\beta^2x^2y^2] = \alpha\beta x^2y^2 [1 - \alpha - \beta]$$

$$\text{trace} =_{\text{sgn}} [\alpha(\alpha-1)y^2 + \beta(\beta-1)x^2]$$

$$\det > 0 \Leftrightarrow \alpha + \beta < 1, \text{ trace} < 0 \Leftrightarrow \alpha < 1 \text{ \& } \beta < 1.$$

Given $\alpha > 0$, $\beta > 0$, condition on trace is implied by condition on det, namely $\alpha + \beta < 1$.

- (iii) $h(x, y) = -2x^2 + kxy - 3y^2$

Hessian:

$$\begin{pmatrix} -4 & k \\ k & -6 \end{pmatrix}$$

$$\det = 24 - k^2, \text{ trace} = -10$$

$\det > 0 \Leftrightarrow k^2 < 24$, $\text{trace} < 0$, so, when $k \in (-\sqrt{24}, \sqrt{24})$, H is negative definite and h is concave.

- (c) (i) Marginal utility tends to infinity as x or y tend to 0.
(ii) Lagrangian $\mathcal{L} = \delta \ln x + (1 - \delta) \ln y + \lambda(m - p_x x - p_y y)$.
K-T FOCs:

$$\delta x^{-1} = \lambda p_x$$

$$(1 - \delta)y^{-1} = \lambda p_y$$

$$\lambda \geq 0, \quad p_x x + p_y y \leq m, \quad \lambda(m - p_x x - p_y y) = 0$$

- (iii) $p_x x^* = \delta m$, $p_y y^* = (1 - \delta)m$, so $x^* = \delta m/p_x$, $y^* = (1 - \delta)m/p_y$

For remaining parts, see diagram at end with budget sets $\mathcal{E}$ optima

- (iv) Under (F):

$$x_A^* = \frac{\frac{4}{5} \cdot 100}{2} = 40, \quad y_A^* = \frac{\frac{1}{5} \cdot 100}{1} = 20; \quad x_B^* = \frac{\frac{2}{5} \cdot 100}{2} = 20, \quad y_B^* = \frac{\frac{3}{5} \cdot 100}{1} = 60.$$

Under (T):

$$x_A^* = \frac{\frac{4}{5} \cdot 180}{2} = 72, \quad y_A^* = \frac{\frac{1}{5} \cdot 180}{3} = 12; \quad x_B^* = \frac{\frac{2}{5} \cdot 180}{2} = 36, \quad y_B^* = \frac{\frac{3}{5} \cdot 180}{3} = 36.$$

A is better off under (T) since the (F)-optimal bundle is affordable but not chosen.

For B , the (F)-optimal bundle is not affordable under (T), and vice versa; direct calculation shows that B is better off under (F).

- (v) A is worse off by imposing (F) as well as (T) since the (F)-optimal bundle (40, 20) is now A 's optimal bundle – the (T)-constraint doesn't bind.

B is also worse off – both constraints bind and B 's optimum is at the kink.

- (vi) A is not made worse off by imposing (T) as well as (F) since the (F)-optimal bundle (40, 20) is still affordable.

B is made worse off by imposing (T) as well as (F) since and B 's optimum moves to the kink.

To make B as well off as they were when only the (F)-constraint was in place, we need to shift the (T)-constraint out so that the kink is at B 's (F)-optimal bundle (20, 60), by making that bundle just affordable under the (T)-constraint: $2 \times 20 + 3 \times 60 = 220$.

Such a relaxation would not benefit A since the stricter (T)-constraint wasn't binding anyway.

3. (a) The function $\mathbf{z}(\cdot)$ has the GSP if for any prices $\mathbf{p}, \mathbf{p}'$ such that for some l $p'_l > p_l$, and $p'_k = p_k$ for all $k \neq l$, then $z_k(p') > z_k(p)$.

If $\mathbf{z}(\cdot)$ is differentiable then this is equivalent to $\frac{\partial z_k(\mathbf{p})}{\partial p_l} > 0$.

- (b) (i) $z_l^a(\mathbf{p}) = \frac{\mathbf{p}\omega^a}{p_l} - \omega_l^a$, $z_k^a(\mathbf{p}) = -\omega_k^a$ for all $k \neq l$.
(ii) Denote by A_k the set of consumers spending all their wealth on good k .

Then the aggregate excess demand of this economy is

$$\mathbf{Z}_k(\mathbf{p}) = \sum_{a \in A_k} \frac{\mathbf{p}\omega^a}{p_k} - \sum_{a \in A} w_k^a$$

and so $\frac{\partial \mathbf{Z}_k(\mathbf{p})}{\partial p_l} = \sum_{a \in A_k} \frac{\omega_l^a}{p_k} > 0$.

- (c) Suppose that $\mathbf{p} \neq \mathbf{p}'$ with $\mathbf{Z}(\mathbf{p}) = 0$, $\mathbf{Z}(\mathbf{p}') \neq 0$.

Since $\mathbf{Z}(\cdot)$ is HoD 0, we can re-scale $\mathbf{p}'$ such that $p_k = p'_k$ and $p_l < p'_l$, and hence by GSP, $Z_k(\mathbf{p}') > Z_k(\mathbf{p}) = 0$.

By Walras' Law $Z_l(\mathbf{p}') = -\frac{p'_k}{p'_l} Z_k(\mathbf{p}') = -\frac{p_k}{p'_l} Z_k(\mathbf{p}')$. Therefore:

$$\mathbf{p}\mathbf{Z}(\mathbf{p}') = p_k Z_k(\mathbf{p}') + p_l Z_l(\mathbf{p}') = (p_k - \frac{p_l p_k}{p'_l}) Z_k(\mathbf{p}') = p_k (1 - \frac{p_l}{p'_l}) Z_k(\mathbf{p}') > 0$$

- (d) Suppose that $\mathbf{p}^*$ and $\mathbf{p}^{**}$ are equilibrium prices and consider $\mathbf{p} = t\mathbf{p}^* + (1-t)\mathbf{p}^{**}$ with $t \in (0, 1)$.

Suppose that $\mathbf{p}$ is not an equilibrium price so that $\mathbf{Z}(\mathbf{p}) \neq 0$, then by the WAE, $\mathbf{p}^*\mathbf{Z}(\mathbf{p}) > 0$ and $\mathbf{p}^{**}\mathbf{Z}(\mathbf{p}) > 0$. But then

$$\mathbf{p}\mathbf{Z}(\mathbf{p}) = t\mathbf{p}^*\mathbf{Z}(\mathbf{p}) + (1-t)\mathbf{p}^{**}\mathbf{Z}(\mathbf{p}) > 0$$

which contradicts Walras' Law.

Therefore the set of equilibrium prices is convex and since generically there are finitely many equilibria, this implies the equilibrium is unique.

4. (a) (i) Points on the straight line joining the 2-vertex with the midpoint of the opposite side (which connects the 1- & 3-vertices) all have a mean of 2.
Other lines whose points have the same mean as each other are parallel to that line.

- (ii) Given a lottery L , we know from the previous part where are the lotteries with the same mean as L .

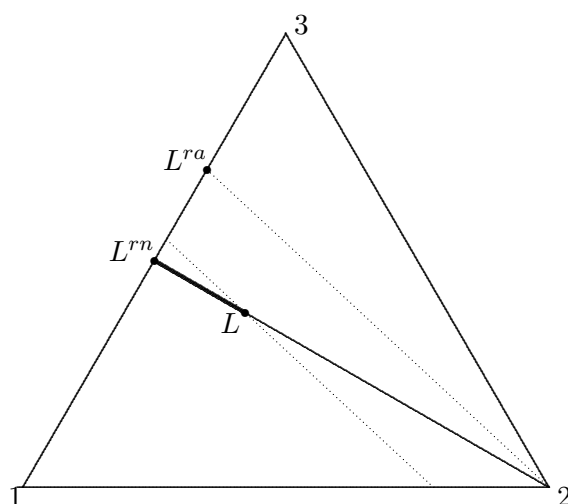
Spreads of L have a higher probability of 1 & 3, and a lower probability of 2, so are closer than L to the side that connects the 1- & 3-vertices.

- (iii) Indifference curves of an expected-utility maximiser are parallel straight lines. A risk-neutral expected-utility maximiser cares only about the mean outcome. Therefore the indifference curve of such an agent passing through the point representing L will be one of those described in the first part.

An indifference curve of a risk-averse agent joins the 2-vertex to a point on the opposite side closer to the 3-vertex than it is to the 1-vertex.

The indifference curve through L of this risk-averse agent is a straight line parallel to that.

As such, its slope is larger than the region/line described in part (a.ii).



Risk neutral:

indifference between 2 and L^{rn}

Risk averse:

indifference between 2 and L^{ra}

Mean-preserving spreads:

on straight line joining L and L^{rn}

- (b) (i) Expenditure = Income, therefore $c \cdot 1 = x\tilde{w}$, and $V(x) = E[u(x\tilde{w}) - x]$.

FOC: $V'(x) = E[\tilde{w} u'(x\tilde{w}) - 1] = 0$.

SOC: $V''(x) = E[\tilde{w}^2 u''(x\tilde{w})] < 0$, since $\tilde{w}^2 > 0$ and $u''(x\tilde{w}) < 0$.

- (ii) $u(c) = 2c^{1/2}$, so $u'(c) = c^{-1/2}$, and the FOC becomes

$$\begin{aligned} E[\tilde{w} (x\tilde{w})^{-1/2} - 1] &= 0 \\ x^{-1/2} E[\tilde{w}^{1/2}] &= 1 \\ x^* &= (E[\tilde{w}^{1/2}])^2 \end{aligned}$$

$f(y) = y^{1/2}$ is a concave function, so $E[\tilde{w}^{1/2}]$ decreases as $\tilde{w}$ becomes riskier.

Therefore x^* falls, i.e. the person works less.

5. (a) $E[u|e=0] = \frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 9 - 0 = 3$

$E[u|e=1] = \frac{1}{3} \cdot 0 + \frac{2}{3} \cdot 9 - 2 = 6 - 2 = 4.$

$\therefore$ agent will exert effort & $EU = 4.$

(b) To fully insure the agent when $e=0$, P. must offer $w_0 = 4^2 = 16.$

$\therefore E[\pi|e=0] - w_0 = \frac{2}{3} \cdot 0 + \frac{1}{3} \cdot 81 - 16 = 27 - 16 = 11.$

To fully insure the agent when $e=1$, P. must offer $w_1 = (4+2)^2 = 36.$

$\therefore E[\pi|e=1] - w_1 = \frac{1}{3} \cdot 0 + \frac{2}{3} \cdot 81 - 36 = 54 - 36 = 18.$

$\therefore \underline{e^* = 1}$, & the agent must end up with 36.

$\therefore s = 81 - 36 = \underline{45}$, & $f = 36 - 0 = \underline{36}.$

(c) To induce high (unobservable) effort, P. will offer the agent utility v_L after a failure & v_H after a success so that the Incentive Compatibility constraint binds, and the Individual Rationality (Participation) constraint binds.

(IC): $\frac{1}{3}v_L + \frac{2}{3}v_H - 2 = \frac{2}{3}v_L + \frac{1}{3}v_H - 0$

(IR): $\frac{1}{3}v_L + \frac{2}{3}v_H - 2 = 4$

$\therefore$ (IC) becomes $\frac{2}{3}v_L + \frac{1}{3}v_H = 4.$

& now $2v_L + v_H = 12$ & $v_L + 2v_H = 18.$

$\therefore v_H + v_L = 10$ & $v_H - v_L = 6$, so $v_H = 8$ & $v_L = 2$

$\therefore$ the agent gets $w_H = 64$ after success, $w_L = 4$ after failure.

So s would be $81 - 64 = \underline{17}$, & f would be $4 - 0 = \underline{4}.$

$E[w|e=1] = \frac{1}{3} \cdot 4 + \frac{2}{3} \cdot 64 = 44$, so agency cost is $44 - 36 = 8$

& expected net profit is now $54 - 44 = 10$

But by fully insuring the agent (who would then exert no effort), P.'s expected net profit is 11 (from part (b)).

So $s = 81 - 16 = \underline{65}$, & $f = 16 - 0 = \underline{16}$; A. exerts no effort

(d) With unobservable effort, agency cost > increased exp. net π .

6. (a) Denoting by q the price of γ and z^A the portfolio demand for agent A , the budget constraint for the agent A is $x_0 = 1 - qz^A$, $x_1 = x_2 = z^A$. Substituting in the utility function, the agent chooses z^A to maximise $\ln(2 - qz^A) + 2\ln(1 + z^A) + \ln(1 + z^A)$. The FOC is

$$\frac{-q}{2 - qz^A} + \frac{2}{1 + z^A} + \frac{1}{1 + z^A} = 0$$

Solving this equation we get $z^A = \frac{6-q}{4q}$.

Analogously, agent B chooses z^B to maximise $\ln(1 - qz^B) + \ln(2 + z^B) + 2\ln(2 + z^B)$. The FOC is

$$\frac{-q}{1 - qz^B} + \frac{1}{2 + z^B} + \frac{2}{2 + z^B} = 0$$

Solving this equation we get $z^B = \frac{3-2q}{4q}$.

The market clearing condition in equilibrium implies $z^A = -z^B$, so this gives $q^* = 3$.

The equilibrium portfolio and allocations are:

$$z^A = \frac{1}{4} = -z^B, \quad x^A = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right), \quad x^B = \left(\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right).$$

The vectors of marginal utilities for the agents are:

$$MU^A = \left(\frac{4}{5}, \frac{8}{5}, \frac{4}{5}\right), \quad MU^B = \left(\frac{4}{7}, \frac{4}{7}, \frac{8}{7}\right).$$

These are not collinear and so the allocation is not Pareto efficient.

The utilities are strongly monotonic so the allocation is constrained Pareto efficient.

- (b) No one in this economy can supply the asset as no one has positive endowment in both states. Hence this asset cannot be offered in equilibrium.

This means that at the equilibrium prices no one should be willing to buy.

Given the symmetry of the utility functions it is enough to check whether B will be willing to buy the asset at the proposed price.

The portfolio demand for B is given by the FOC:

$$\frac{-q}{3 - qz} + \frac{1}{1 + z} + \frac{2}{2 + z} = 0$$

Replacing q by 3 the solutions of this quadratic equation are $\left(\frac{\sqrt{3}-1}{2}, -\frac{\sqrt{3}+1}{2}\right)$; the second one is smaller than -1 and cannot be a demand (Agent B 's utility would be undefined), so the solution is $z^B = \frac{\sqrt{3}-1}{2} > 0$.

So there is demand for γ that cannot be met and this is not an equilibrium price.

- (c) In equilibrium the only one that could sell asset α is A .

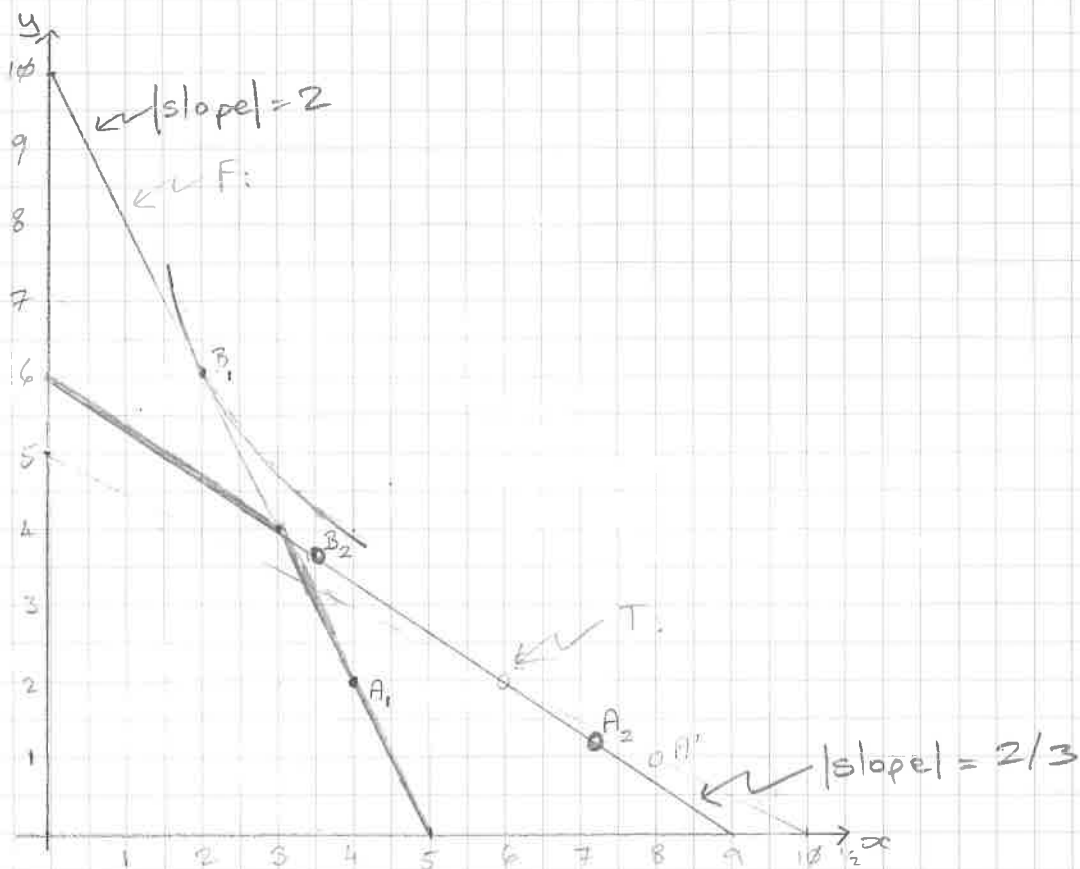
He/she will do so in strictly positive quantities as B will have a positive demand for it, and will be able to sell asset γ to A .

The equilibrium price for γ should be higher than for α .

If not there would be an arbitrage opportunity:

if $q_\gamma < q_\alpha$, then an agent could buy one unit of γ and sell one unit of α with negative cost and make strictly positive profits.

In this situation we have complete markets and the equilibrium allocation will be efficient.



$$\left. \begin{aligned} u_a &= 4 \ln x + \ln y \\ u_b &= 2 \ln x + 3 \ln y \end{aligned} \right\}$$

$$2x + y \leq 100 \quad BC_1$$

$$2x + 3y \leq 180 \quad BC_2$$

Increase 180 to 220
makes B as well off as before